



ASSIGNMENT-4: MULTI-MODAL HUMAN DETECTION SYSTEM



MUHAMMAD SAQLAIN TAHIR
BSSE23050
ZUNAIRA ABDUL AZIZ
BSSE23058
FAIQA ARSHAD
BSSE23028

SECTION A

Group Member	Work Done
Faiqa Arshad (BSSE23028)	4.1 Module 1: Lips Detection (33%)
Muhammad Saqlain Tahir (BSSE23050)	4.2 Module 2: Eyes Detection 4.3 Module 3: Face Detection (34%)
Zunaira Abdul Aziz (BSSE23058)	4.4 Module 4: Hand Detection 5. Code Structure Requirements (33%)

Description of each Members work:

BSSE23028 Faiqa Arshad:

In my contribution to this project, I focused on the design and implementation of Module 1: Lips Detection, ensuring it met all technical requirements for real-time interaction. I developed the logic to track lip landmarks using MediaPipe FaceMesh, which allowed the system to accurately calculate the Mouth Aspect Ratio (MAR). To enhance usability, I implemented Smile Detection by comparing lip-corner positions relative to the mouth center and integrated a Lip-Sync Counter that utilizes hysteresis with two distinct thresholds (0.55 and 0.35) to prevent false triggers. Furthermore, I ensured the system supports multi-modal inputs, including live webcam feeds, video file uploads, and static images. I specifically programmed the video controls to allow for play, pause, and replay functionality, alongside real-time visual alerts like the red frame border when a yawn is detected. My work balanced technical accuracy with HCI principles to create a responsive and user-friendly interface.

BSSE23050 Saqlain Tahir:

For this group project, I managed the core development of both Module 2 which is Eyes Detection and Module 3 which is Face Detection. I aimed to develop a responsive system which could recognize human face gestures accurately. For the eyes module, I implemented a robust eye aspect ratio (EAR) via MediaPipe landmarks. I created the logic to count blinks, and furthermore the drowsiness alert also. I created a temporary buffer so that if a user's eyes remain closed for more than 20 frames, a prominent red warning appears. This mimics a real-world driver monitoring safety feature.

Within the Face Detection unit, I implemented DeepFace and built a head pose estimation algorithm for your detection of where the person is looking and if the head is tilted up or down. In order to maintain smoothness of the interface, an emotion history log was added to the project, which analyses the last five seconds of data to display a person's "recent mood". This was added to filter out random glitches. I also made sure these modules are working on different inputs like live webcam, video files, and static images. I learned to convert rough raw landmark data into human-computer interactions which ultimately feel human-centered and more intuitive for the user.

BSSE23058 Zunaira Abdul Aziz:

For this group project, I was responsible for two main parts: the Hand Detection module and the overall Code Structure. For the hand detection, I used the MediaPipe library to track the user's hand movements on the screen. I wrote the code to count exactly how many fingers the user is holding up. Using that information, I added a feature to recognize specific hand signs, such as a Fist, a Peace sign, an Open Hand, and a Thumbs Up. To make the project more fun, I also built a Gesture Game. In this game, a random gesture pops up on the screen. If the user successfully copies that gesture and holds it for one full second, a timer bar fills up and they earn a point.

Along with the camera features, I was also in charge of organizing our team's work. I combined everyone's code into one single Python file, exactly as the assignment required. I built the main menu system so the user can easily choose if they want to use a live webcam, a video file, or an image. I also made sure that pressing the ESC key always safely takes the user back to the main menu. Finally, I checked that all of our code was clean, making sure we used proper function notes and avoided using global variables. My work helped ensure that our final project was organized, easy to use, and completely followed all the rules.